CM0081 Formal Languages and Automata
§ 1.4 Formal Proofs

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Preliminaries

Conventions

- The number and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook (Hopcroft, Motwani and Ullman 2007).
- The natural numbers include the zero, that is, $\mathbb{N} = \{0, 1, 2, \dots\}$.
- The power set of a set A, that is, the set of its subsets, is denoted by $\mathcal{P}A$.

Proofs by Contradiction and Proofs of Negations

Proof by contradiction
(or *reductio ad absurdum*)

$$\dfrac{\begin{matrix}[\neg\beta] \\ \vdots \\ \bot\end{matrix}}{\beta}$$

Proof of negation (Bauer 2017)

$$\dfrac{\begin{matrix}[\beta] \\ \vdots \\ \bot\end{matrix}}{\neg\beta}$$

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Justifications

$$\dfrac{\dfrac{\dfrac{\begin{matrix}[\neg\beta] \\ \vdots \\ \bot\end{matrix}}{\neg\beta \to \bot}\ \text{(conditional proof)}}{\neg\neg\beta}\ (\neg\alpha := \alpha \to \bot)}{\beta}\ (\vdash \neg\neg\alpha \to \alpha)$$

$$\dfrac{\dfrac{\begin{matrix}[\beta] \\ \vdots \\ \bot\end{matrix}}{\beta \to \bot}\ \text{(conditional proof)}}{\neg\beta}\ (\neg\alpha := \alpha \to \bot)$$

Inductive Proofs: Mathematical Induction

The induction principle

Let $S(n)$ be a property about natural numbers. If

(i) we prove $S(i)$ (basis step) and

(ii) we prove that for all natural number $n \geq i$, $S(n)$ implies $S(n+1)$ (inductive step),

then we may conclude $S(n)$ for all $n \geq i$.

Inductive Proofs: Structural Induction

The structural induction principle

Let $S(X)$ be a property about structures X that are defined by some recursive/inductive definition. If

(i) we prove $S(X)$ for the basis structure(s) of X (basis step) and

(ii) given a structure X whose recursive/inductive definition says it is formed from $Y_1, \ldots, Y_k$, we prove $S(X)$ assuming that the properties $S(Y_1), \ldots, S(Y_k)$ hold (inductive step),

then $S(X)$ is true for all X.

Inductive Proofs: Mutual Induction

Example

Given the functions $f, g, h : \mathbb{N} \to \mathbb{N}$,

$$f(0) = 0, \qquad g(0) = 1, \qquad h(0) = 0,$$
$$f(n+1) = g(n), \qquad g(n+1) = f(n), \qquad h(n+1) = 1 - h(n),$$

and properties R, S, T,

$$R(n)\colon h(n) = 1 - g(n), \qquad S(n)\colon h(n) = f(n), \qquad T(n) : S(n) \wedge R(n),$$

Inductive Proofs: Mutual Induction

Example

Given the functions $f, g, h : \mathbb{N} \to \mathbb{N}$,

$$\begin{aligned} f(0) &= 0, & g(0) &= 1, & h(0) &= 0, \\ f(n+1) &= g(n), & g(n+1) &= f(n), & h(n+1) &= 1 - h(n), \end{aligned}$$

and properties R, S, T,

$$R(n) \colon h(n) = 1 - g(n), \qquad S(n) \colon h(n) = f(n), \qquad T(n) : S(n) \wedge R(n),$$

1. To prove $(\forall n) R(n)$ (impossible!)
2. To prove $(\forall n) S(n)$ (impossible!)

Inductive Proofs: Mutual Induction

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$$R(n) \colon h(n) = 1 - g(n), \qquad S(n) \colon h(n) = f(n), \qquad T(n) : S(n) \wedge R(n),$$

1. To prove $(\forall n)R(n)$ (impossible!)
2. To prove $(\forall n)S(n)$ (impossible!)
3. To prove $(\forall n)T(n)$ (by mutual induction)

(continued on next slide)

Inductive Proofs: Mutual Induction

Proof

- Basis step $T(0)$. Easy.

Inductive Proofs: Mutual Induction

Proof

- Basis step $T(0)$. Easy.
- Induction step $T(n) \Rightarrow T(n+1)$:

$$
\begin{aligned}
S(n): \quad h(n+1) &= 1 - h(n) && \text{(def. of } h\text{)} \\
&= g(n) && \text{(IH } R(n)\text{)} \\
&= f(n+1) && \text{(def. of } f\text{)}
\end{aligned}
$$

$$
\begin{aligned}
R(n): \quad h(n+1) &= 1 - h(n) && \text{(def. of } h\text{)} \\
&= 1 - f(n) && \text{(IH } S(n)\text{)} \\
&= 1 - g(n+1) && \text{(def. of } g\text{)}
\end{aligned}
$$

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References

Andrej Bauer (2017). Five States of Accepting Constructive Mathematics. Bulletin of the American Mathematical Society 54.3, pp. 481–498. DOI: 10.1090/bull/1556 (cit. on pp. 3, 4).

John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman [1979] (2007). Introduction to Automata Theory, Languages, and Computation. 3rd ed. Pearson Education (cit. on p. 2).